<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks Available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>3.</td><td>(a)</td><td></td><td></td><td>
<table>
<tr><th></th><th>Geosynchronous satellite</th><th>Geostationary satellite</th></tr>
<tr><td>Stays above the same point on Earth at all times</td><td>(×)</td><td>✓</td></tr>
<tr><td>Orbits Earth once in 24 hours</td><td>✓</td><td>✓</td></tr>
<tr><td>Orbits above the equator</td><td>×</td><td>✓</td></tr>
<tr><td>Must pass over the North pole and South pole</td><td>×</td><td>×</td></tr>
</table>

Award 1 mark for each correct row</td><td>4</td><td></td><td></td><td>4</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td></td><td>Substitution into: $v = f\lambda$ so $3 \times 10^8 = f \times 0.08$ (1)
Rearrangement: $f = \frac{v}{\lambda} = \frac{3 \times 10^8}{0.08}$ (1)
Frequency = 3.75×10^9 (1)
Answer of 3.75×10^n where n is not = to 9 award 2 marks
Unit = Hz (1) accept Hertz</td><td>1

1</td><td>1
1</td><td></td><td>4</td><td>3</td><td></td></tr>
</table>

Question				Marking details	Marks Available AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Substitution into: $\text{speed} = \frac{\text{distance}}{\text{time}}$ so $3 \times 10^8 = \frac{\text{distance}}{0.24}$ (1) Distance = 7.2×10^7 [m] (1) Distance to satellite = half calculated distance = 3.6×10^7 [m] (1) Answer of 7.2×10^n where n is not = to 7 award 1 mark Answer of 14.4×10^7 [m] award 2 marks **OR:** Halve time to 0.12 s (1) Substitution into: $\text{speed} = \frac{\text{distance}}{\text{time}}$ so $3 \times 10^8 = \frac{\text{distance}}{0.12}$ (1) = 3.6×10^7 [m] (1) Answer of 3.6×10^n where n is not = to 7 award 2 marks	1	1 1		3	3	
				Question 3 total	**7**	**4**	**0**	**11**	**6**	**0**